

Learned-Only Multi-Agent Control of a Relay-Drone Network for Disaster Relief Delivery: What a Set-Based Policy Wins and Where a Geometric Rule Still Leads

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Abstract

In a disaster zone without communication infrastructure, a fleet of homogeneous drones must deliver relief supplies while every aircraft stays connected to the control centre through a multi-hop relay chain formed by the fleet itself. A drone that holds a relay position cannot deliver, so the fleet trades throughput for reach at every decision. We study the learned-only setting: a single set-based policy, trained once with three to four drones on twenty to eighty destinations, is deployed without retraining on fleet sizes, destination counts and depot positions it has never seen. Connectivity is a hard constraint enforced by a minimal-perturbation projection; a high-level manager assigns each drone a destination, a return or a relay slot every twenty steps, deciding drones

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autoregressively from the farthest inward; the low-level controller is straight-line flight. The manager is trained with discrete soft actor-critic whose entropy target is annealed from $0.6 \log n$ to $0.25 \log n$, regularised toward a geometric relay rule, and selected on a held-out makespan score.

Against the strongest learning-free baseline we could build, a geometric relay rule augmented with a deadlock-escape mechanism, the learned policy is evaluated by pairing the two methods on identical instances and testing the paired differences with the Wilcoxon signed-rank test on 200 to 240 rollouts per configuration. The learned policy completes every instance in all three configurations where the rule fails 3 of 240 and 10 of 200. On the unseen five-drone configuration it is faster by 54 steps in paired median ($p < 10^{-4}$); on the training configuration the two are tied (+18, $p = 0.17$); on the unseen three-drone configuration the rule is faster by 40 steps ($p < 10^{-4}$). A high-level decision costs 3.5 ms at four drones, against 600–1200 s for a time-expanded MILP that finds no feasible plan beyond six destinations. We trace the three-drone deficit to relay assignment: the policy stations 0.78 relays where the rule stations 1.39, and overriding only the rule’s choice of *which* drone relays recovers the entire gap. Five training-side interventions aimed at that decision, spanning imitation weighting, reward credit for relays, checkpoint selection and per-drone critics, all failed, and we report them.

1 Introduction

Relief delivery after a disaster is a routing problem with an unusual coupling. Communication infrastructure is gone, so each drone can talk to the control centre only within a direct range R , or through other drones that forward its traffic. Every aircraft must remain connected at all times, for live video during unloading and for tracking; a drone that stations itself as a relay extends the fleet’s reach at the cost of its own deliveries. The fleet therefore makes, continuously, a joint decision about who delivers and who relays, and the makespan of serving all demand depends on that decision as much as on the routes.

Exact optimisation does not scale here. A time-expanded mixed-integer model that enforces n -hop connectivity in every period has, for three drones and ten destinations on a 98-metre lattice, about 11,000 binaries and finds no feasible plan in ninety minutes (Section 8). A geometric rule that places relays at fixed fractions of the line from the depot to the lead drone is fast and, once a deadlock-escape mechanism is added, completes almost every instance. It is the baseline any learned method must beat, and this paper reports carefully where a learned policy does and does not.

Our contribution is a learned-only controller and an honest account of it. The controller is a set-based, drone-count-agnostic manager trained once and deployed on unseen fleet sizes and depot positions. Its wins are unambiguous on completion rate, on the unseen five-drone configuration, and on inference cost. Its loss on the three-drone configuration is measured, diagnosed to a single decision, and shown to resist five plausible fixes. We also document two methodological corrections that changed our conclusions: that comparing makespan only over the episodes each method completed favours the method that gives up on hard instances, and that forty-rollout samples cannot support claims about tails or about differences of thirty steps.

2 Related Work

Routing with drones has been studied since the flying-sidekick problem of Murray and Chu (2015) and its multi-drone extensions (Murray & Raj 2020; Agatz et al. 2018; Dorling et al. 2017); none of these models communication. UAV relay positioning is its own literature (Zeng, Zhang & Lim 2016; Yanmaz 2022; Ding et al. 2022), but there the relays are the mission, not a role the delivery fleet takes on and gives up. Multi-UAV exploration under communication limits (Cesare et al. 2015) is the closest in spirit. On the learning side we build on soft actor-critic (Haarnoja et al. 2018), centralised critics for decentralised execution (Lowe et al. 2017; Foerster et al. 2018), value decomposition (Sunehag et al. 2018; Rashid

et al. 2018), options as a semi-Markov decision process (Sutton, Precup & Singh 1999), invalid-action masking (Huang & Ontañón 2022) and safe-action projection (Dalal et al. 2018). Learned solvers for min-max multi-vehicle routing (Park, Kwon & Park 2023) show set-based policies generalising across problem sizes, which is the property we need across fleet sizes.

3 Problem Statement

A control centre sits at $c \in [0, 1000]^2$ and M destinations $d_1, \dots, d_M$ lie within $0.9 NR$ of it, where N is the number of drones and $R = 300$ m the communication range. Each drone flies at $v = 13.89$ m/s, carries $Q = 5$ units, unloads one unit per destination in $S = 10$ steps, and reloads to Q at the centre in S steps; the interaction radius is 25 m and one step is one second. The fleet must stay connected: at every step the graph with the centre and the drones as vertices and edges between vertices within R must be connected. The objective is the makespan, the step at which the last destination is served. We call the $N = 4$, $M = 50$, fixed-depot setting the *standard* configuration; the policy is trained with $N \in \{3, 4\}$, $M \in [20, 80]$ and a random depot, and evaluated in addition on two configurations it never saw, $N = 3$, $M = 30$ and $N = 5$, $M = 50$, both with random depots.

4 Method

4.1 Connectivity as a hard constraint

A reward penalty for disconnection punishes situations no action can prevent: a drone that is stationary while unloading is stranded when a teammate departs. Connectivity is therefore enforced by projection. Each commanded displacement is scaled by the largest factor in $[0, 1]$ that keeps the fleet connected, with ties between drones broken at random. The scaled-away distance is recorded as a *blocked* drone-step; blocked steps turn out to be

the diagnostic signature of under-relaying (Section 7). No connectivity loss occurred in any training episode or evaluation rollout reported here.

4.2 Hierarchy and the manager’s action space

A manager acts every 20 steps, or as soon as a drone’s goal becomes invalid, and assigns each drone one of $K = 7$ discrete options: one of five candidate destinations (the three nearest unserved ones and the two farthest from the centre), a return to the centre, or a relay slot. Relay slots lie on the segment from the centre to the lead drone at multiples of $0.9R$; the environment assigns the slot, so the policy’s relay decision is only *whether* a drone relays. The low-level controller is straight-line flight to the goal; a learned worker was tried in earlier versions and was harmful. Between manager decisions the environment is a semi-Markov decision process and the manager’s transitions store the discounted option return.

4.3 Set-based, drone-count-agnostic manager

The manager encodes each drone as a set of entities: its own state (10 features), its five candidates (6 features each), its peers (9 features each, of which four are a one-hot of the peer’s current role), and its relay slot (4 features). Attention pooling over candidates and over peers produces a per-drone embedding from which a pointer head scores the candidates and a fixed head scores return and relay. Nothing in the network depends on N or M , which is what allows a single set of weights to run on fleet sizes it never saw. Invalid options are masked.

Decisions are *autoregressive*: drones are ordered by distance from the centre, farthest first, and after each drone’s option is sampled its new role is written into the peer features seen by the drones decided after it. A formation of one lead and k relays arises only by coincidence under independent per-drone sampling but can be matched deliberately under sequential decisions. The price is N sequential network calls per decision.

4.4 Training

The manager is trained with discrete soft actor-critic. A twin-Q critic with attention across drone embeddings outputs per-drone action values whose mean over drones is regressed on the team return; the actor loss is the standard expectation form. Three ingredients matter.

Entropy target. The conventional target of $0.6 \log n_{\text{valid}}$ holds the temperature at 2–3 and keeps the policy too random to hold a multi-step formation, while that same randomness is the only mechanism by which a deadlocked fleet escapes. Annealing the target linearly from $0.6 \log n$ to $0.25 \log n$ over the first 400 episodes lowered the standard- configuration makespan from 1,418 to the 900s; it is the single largest contributor.

Rule regularisation. A cross-entropy term with weight 2.0 pulls the policy toward the action the geometric rule would take, wherever that action is unmasked. Removing it collapses completion early in training and costs 300–400 steps of makespan.

Configuration randomisation. Training samples N , M and the depot position afresh every episode. Removing it makes the standard configuration 44 steps faster and the unseen three-drone configuration 646 steps slower.

Checkpoints are scored every 50 episodes on 80 held-out instances with a 3,000-step budget by $100f + 40(1 - \bar{t}/3000)$, where f is the completion rate and $\bar{t}$ the mean steps, so that one percentage point of completion equals 75 steps; the earlier weight of 10 made it equal 300 and discarded fast checkpoints for a single missed instance. Training stops after eight consecutive evaluations without improvement or at 1,000 episodes.

5 Baselines and Evaluation Protocol

5.1 Five learning-free baselines

We compare against a greedy assignment with no relays, a geometric relay rule, the same rule with a deadlock-escape mechanism, and two fixed-relay policies that permanently dedicate

one or two drones. The geometric rule stations relays along the depot-to-lead segment and is deterministic; it deadlocks on 38% of standard instances. The escape mechanism swaps the lead drone whenever the fleet has been stalled for 60 steps and holds the new assignment for 40 steps, which lifts completion to 97–100%. Table 1 reports all five on all three configurations; the augmented rule is the only credible baseline and every comparison below is against it.

Table 1: Learning-free baselines: completion rate and mean makespan over completed episodes. Standard configuration 120 rollouts, unseen configurations 40.

Method	4 drones / 50 dest.	3 / 30 (unseen)	5 / 50 (unseen)
Greedy, no relay	0%	0%	25% · 516
One fixed relay	0%	2.5%	40% · 544
Two fixed relays	3.3% · 914	52.5% · 875	65% · 611
Geometric relay rule	61.7% · 810	95% · 581	92.5% · 598
Rule + deadlock escape	98.3% · 887	100% · 607	92.5% · 623
Learned policy (ours)	100% · 911	100% · 689	100% · 593

5.2 Paired comparison

Averaging makespan over the episodes each method completed is biased: the hard instances a method gives up on drop out of its average and stay in the other’s. A second problem is that the learned policy has a heavy tail; one 8,336-step rollout moves a 40-instance mean by 110 steps, and the same weights measured three times on the standard configuration gave means of 899, 985 and 950. We therefore roll both methods out on the same instances with the same repetition index, keep the pairs both completed, and report the median paired difference, the win–loss count and the Wilcoxon signed-rank p -value, with 240 rollouts on the standard configuration and 200 on each unseen one. Inference samples from the policy; the greedy argmax has the same median and a far heavier tail (maximum 8,482 against 2,538 on 120 rollouts), which confirms that sampling is the deadlock-escape mechanism. All rollouts have a 10,000-step budget and no early termination.

6 Results

6.1 Three configurations

Table 2: Learned policy against the rule with escape on identical instances. Positive paired difference means the learned policy is slower. p from the Wilcoxon signed-rank test.

Configuration	Rollouts	Completed rule / ours	Median rule / ours	Paired diff.	Wins-losses	p
4 / 50, seen	240	237 / 240	866 / 879	+18	109–127	0.17
3 / 30, unseen	200	200 / 200	572 / 617	+40	73–127	$< 10^{-4}$
5 / 50, unseen	200	190 / 200	570 / 532	− 54	141–45	$< 10^{-4}$

The learned policy completes every instance in every configuration; the rule leaves 3 of 240 and 10 of 200 unfinished within the budget. On the unseen five-drone configuration the learned policy is 54 steps faster in paired median and wins 141 pairs of 186. On the training configuration the two are indistinguishable: two independent 120-rollout measurements gave paired medians of +24 and +17, neither significant, and the pooled p is 0.17. On the unseen three-drone configuration the rule is 40 steps faster and the difference is significant. Figure 1 shows the full distributions.

6.2 Tails

The direction of the tail follows the direction of the median. Where drones are scarce the learned policy’s worst case is worse than the rule’s; where they are plentiful it is better.

Table 3: Standard deviation and maximum of completed makespan.

Configuration (rollouts)	Rule sd / max	Ours sd / max
4 / 50 (240)	188 / 2,014	521 / 8,336
3 / 30 (200)	172 / 1,447	382 / 5,395
5 / 50 (200)	150 / 1,734	111 / 1,101

The 8,336-step event is rare: one in 960 standard rollouts exceeded 2,500 steps, and the same instance run twice gave 702 and 8,336, so it is a stochastic stall, not an instance property. An inference-side intervention that resamples the policy at temperature 5 whenever

Paired comparison on identical instances (budget 10,000 steps, sampling inference)

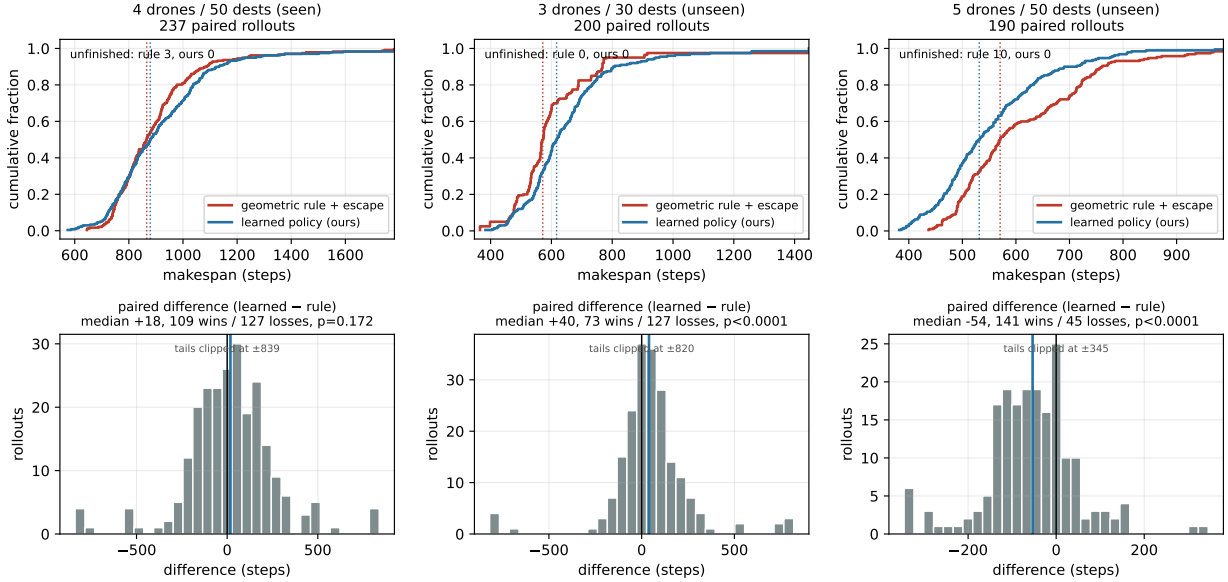


Figure 1: Cumulative makespan distributions (top, dotted lines at medians) and paired differences (bottom) for the three configurations.

the fleet has been stalled for 120 steps cuts the 600-rollout maximum from 2,361 to 1,541 and the share of rollouts above 1,500 steps from 1.2% to 0.3% at no median cost (paired -7 , $p = 0.43$); the tail reduction itself is not yet significant (Fisher $p = 0.18$).

6.3 Inference cost

A high-level decision costs 0.08ms for the rule, 1.52ms for the learned policy without autoregression and 3.52ms with it, at four drones on a CPU; the autoregressive cost grows at 0.93ms per additional drone. A 900-step episode spends 0.158s in the manager. The time-expanded MILP of Section 8 needs 600–1,200s on instances with four to six destinations.

6.4 Ablations

Each component was removed in turn with two seeds. Because the same configuration trained four times spans 919–979 steps on the standard configuration, only differences above 70–80 steps between two-seed means are meaningful; those are in bold.

Table 4: Ablation: completion and mean makespan, two seeds, difference from the full method in parentheses.

Removed	4 / 50	3 / 30	5 / 50
Nothing (full method)	120/120 · 927	40/40 · 755	40/40 · 574
Residual penalty	120/120 · 899 (−28)	40/40 · 718 (−37)	40/40 · 582 (+8)
Autoregression	120/120 · 1,022 (+ 94)	40/40 · 710 (−46)	40/40 · 607 (+33)
Entropy annealing	120/120 · 1,056 (+ 128)	40/40 · 774 (+19)	40/40 · 635 (+61)
Config. randomisation	120/120 · 883 (−44)	40/40 · 1,402 (+ 646)	40/40 · 642 (+68)
Rule regularisation	118/120 · 1,246 (+ 320)	39/40 · 1,163 (+ 408)	40/40 · 666 (+ 92)

Rule regularisation is essential; configuration randomisation is what generalisation costs on the training configuration and buys on the unseen one; entropy annealing and autoregression matter on the standard configuration. The residual penalty is neutral, and a first reading that called it harmful was retracted once four runs of the same configuration were available.

7 Why the Three-Drone Configuration Is Slower

With three drones and destinations up to 810 m away, a far delivery needs two relays, leaving one deliverer. On 40 three-drone instances the rule stations 1.39 relays on average and the learned policy 0.78; the policy sends 1.78 drones to deliver against the rule’s 1.32, and the connectivity projection blocks 1,289 drone-steps against 160. The policy under-relays and is dragged.

A probe isolates the cause. Overriding only the policy’s relay assignments with the rule’s, while leaving its destination choices intact, cuts blocked steps by 60% and recovers the whole gap (paired median −44 against the unmodified policy, versus −43 for the rule itself). Matching only the rule’s relay *count*, with the drones chosen by the policy’s own relay probabilities, collapses the tail (maximum 4,904 to 1,083) but recovers only 14 steps of median. The tail is a matter of how many drones relay; the median is a matter of which.

The information needed for that choice is present: a two-layer classifier predicts the

rule’s per-drone relay decision from the policy’s 41 observation features with 85% accuracy against a 51% majority baseline. What is missing is the training signal. The critic’s greedy policy, the per-drone argmax of its action values, loses to the actor 0–40 with a median makespan of 2,022 against 670, even when actor and critic are taken from the same episode: the critic is trained only through the mean of per-drone values against the team return, so the decomposition is under-determined and the actor’s competence comes from the entropy-regularised policy gradient and the rule term, not from an accurate per-drone value.

7.1 Five interventions that did not close the gap

We report these because each was the natural next step given the diagnosis and each failed. Values are paired medians against the rule on the three-drone configuration, 200 rollouts; the full method sits at +40.

Table 5: Training-side interventions aimed at relay assignment. Two seeds each.

Intervention	Mechanism	3 / 30 paired median
Drop autoregression and residual penalty	simpler policy, 2.3× faster	+62, +132
Rule-regularisation weight $\times 3$ on relay actions	stronger imitation of relaying	+54, +56
Uniform rule-regularisation weight 4.0	stronger imitation overall	+23, +39
Per-drone reward credit to path relays	credit assignment	+36, +58
Mixed holdout (half three-drone) for selection	selection sees the target	+37, +68
Zero-sum credit taken from the deliverer	credit without an incentive to relay	+91, +126

Imitation weighting raised the relay share during training from 0.24 to 0.30 but not toward the rule’s 0.44. Reward credit raised it to 0.36 and reversed the five-drone advantage from -54 to $+60$: a drone parked on a busy path collects credit for every delivery that passes, so relaying became an end in itself; the zero-sum variant removed that incentive and instead starved delivery learning. A control arm that changed only the critic’s target to per-drone rewards, without any credit, was unstable across seeds and lost the five-drone advantage on its own. Selecting checkpoints on a mixed holdout improved the standard configuration (arm mean from $+19$ to 0 , one weight reaching -37) but not the three-drone one, and a sweep over every saved checkpoint of four runs found none closer than 40 steps to the rule on a

fresh seed range. The deficit is a property of the policy, not of selection, reward magnitude or observation.

8 A Time-Expanded MILP as a Reference

For small instances we solve a time-expanded network on a 98.2-metre lattice with periods of 10 steps, single-commodity flow for connectivity in every period and a symmetry-breaking cut; the formulation and its proofs are in a companion document. With three drones the model returns a plan for four and six destinations, which replayed in the simulator beats the rule by 7 and 9 steps, and returns nothing for ten destinations in ninety minutes. Its optimum is a restriction of the continuous problem and not a bound on it. A certified lower bound through a cell relaxation was attempted and abandoned: on the four-destination instance the relaxation’s dual bound after seven minutes is 40 steps, below the trivial analytic bound of 60, because testing communication by the minimum distance between cells inflates a three-hop chain’s reach from 900 m to 1,317 m; keeping that inflation below 10% needs 21-metre cells and roughly 7×10^7 flow variables. The difficulty of the problem lies entirely in the connectivity constraint, and any discretisation coarse enough to solve weakens exactly that constraint.

9 Conclusion

A single set-based policy, trained once, completes every instance across fleet sizes and depot positions it never saw, beats a strong geometric rule decisively when drones are plentiful, ties it on the training configuration, and runs three orders of magnitude faster than exact optimisation. It is slower than the rule by 40 steps when drones are scarce, and we have localised that loss to one decision, which drone should relay, that the current critic cannot value because it never conditions on the joint action. A critic that does, in the manner of centralised critics for decentralised execution, was the last structural lever we tried: with

four seeds it was slower than the same recipe without it on all three configurations (paired medians +58, +89 and -27 against +10, +58 and -42), which we attribute to the variance of a sampled joint action in the target and to the autoregressive actor already carrying the information the critic was given. The remaining option is a manager whose output is a team assignment rather than per-drone options, which is a different design. Two methodological points travel beyond this problem: completed-episode averages favour the method that gives up, and forty rollouts cannot support claims about tails.

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